

▲ AEGIS

Alert Enrichment, Grouping & Intelligence Scoring

Threat-intelligence correlation and alert prioritisation, with an evidence trail IBM Bob can be asked about.

Team Blue Vector · Track: AI

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The problem: four consoles, four severity scales, one analyst

Volume

Thousands of alerts a day across SIEM, network sensors, geospatial feeds and written intelligence. Nobody reads them all, so triage is "whatever sorts to the top".

Incomparable severity

The SIEM says critical, the sensor says err, the geo feed says P2, the report says HIGH. None knows the asset or what else is happening.

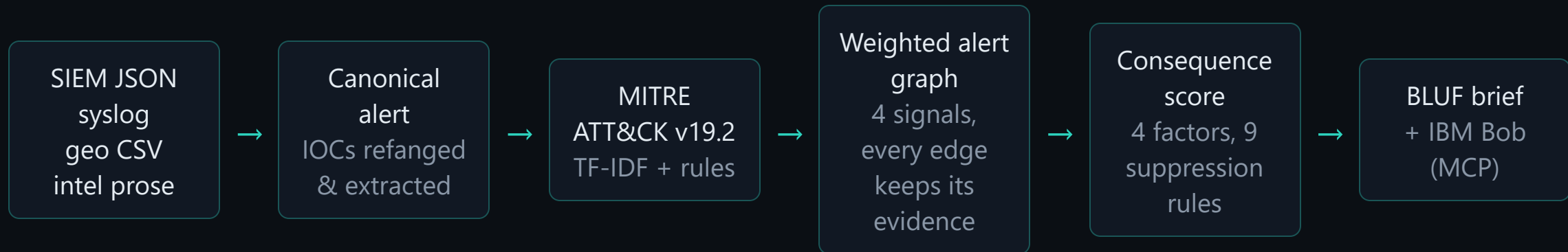
Asymmetric cost

A false positive costs an analyst-hour. A missed intrusion costs the mission. Real intrusions arrive as three unremarkable alerts in three consoles.

[illegible]

What the analyst actually starts the shift with.

AEGLIS: triage as a graph problem with an evidence trail



Four correlation signals per alert pair

- **Shared indicator** ≤ 0.40 — rarity-weighted (IDF over the corpus); a shared SHA-256 beats a shared DNS server
- **Temporal proximity** ≤ 0.20 — exponential decay, no hard window
- **Asset adjacency** ≤ 0.15 — same host, user, subnet, or one alert naming the other's asset
- **Tactic progression** ≤ 0.25 — *directional* kill-chain ordering: Initial Access → Execution scores,

Why it is not "group by source IP"

- Incidents are connected components above a threshold — count emerges from evidence, not a chosen k
- Ordering only chains alerts already tied by an entity; timing alone is coincidence
- Priority is consequence-driven: a medium-confidence hit on a domain controller outranks a certain one on a print server
- Vendor severity is preserved, shown, and never

Architecture: one state, two surfaces

Pipeline (Python 3.10, ~3 s for 521 alerts)

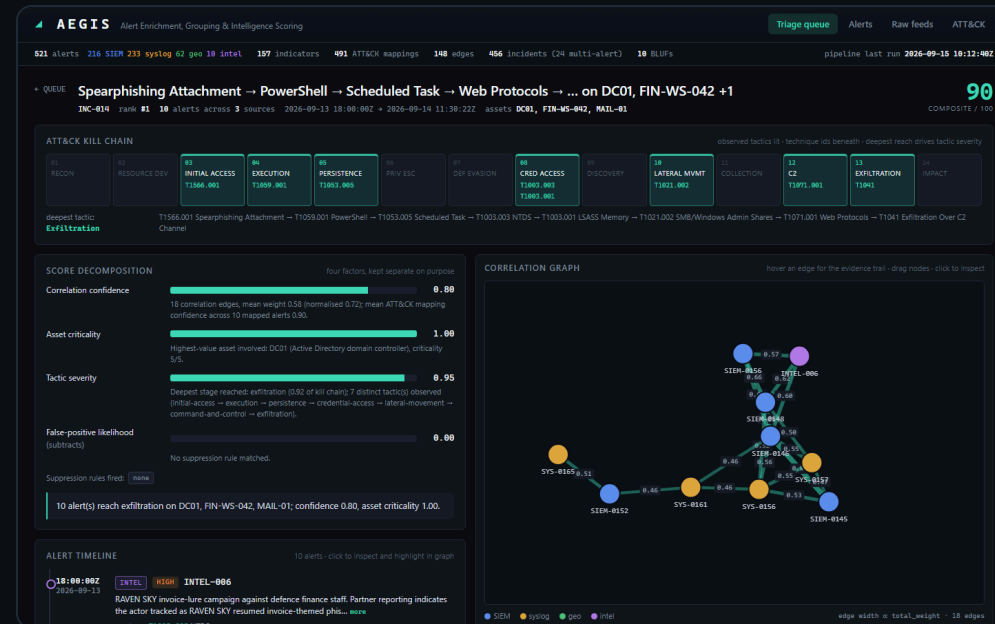
Ingest adapters → SQLite (stdlib) → ATT&CK mapper (numpy TF-IDF over the pinned STIX corpus, 697 techniques) → correlation engine (NetworkX) → prioritiser → BLUF generator (template; watsonx.ai Granite refinement validated against evidence)

Service layer → FastAPI + MCP server

The console and IBM Bob read the same SQLite through the same functions. Neither computes anything, so they cannot disagree.

Console: plain HTML/JS, no build step

Force-directed correlation graph on canvas, evidence on edge hover, kill-chain strip, score decomposition, BLUF panel, "why is this ranked here?"



INC-014: phishing → PowerShell → C2 → DC01 credential theft → exfiltration. 10 alerts, 3 sources, 18 evidential edges.

The closer: "The SIEM marked this CRITICAL. Why is it ranked 136th?"

The screenshot shows the AEGIS SIEM interface. On the left, a table lists alerts with columns for TIME, ALERT, SOURCE, VENDOR SEV., ASSET, TOP TECHNIQUE, and SUMMARY. The main alert shown is SIEM-0121, which is CRITICAL. On the right, a detailed view of this alert is shown, including its timestamp (2026-09-14 06:00:12Z), vendor severity (CRITICAL), asset (DC01), incident (INC-018), and queue position (136 of 456). Below this, a section titled 'WHY IS THIS RANKED HERE?' explains the ranking. It shows that the alert is ranked 136 of 456 based on its vendor severity (CRITICAL) and asset criticality (1.00). A suppression rule is also shown, which is 'Authorised vulnerability scanner' and is fired because the alert is a scan from an authorised range (10.50.1.10) on a scheduled basis. The score breakdown shows a correlation confidence of 0.73, asset criticality of 1.00, tactic severity of 0.49, and a false-positive likelihood of 0.92.

TIME	ALERT	SOURCE	VENDOR SEV.	ASSET	TOP TECHNIQUE	SUMMARY
2026-09-14 23:55:25Z	SYS-0233	SYSLOG	INFO	DNS-01	unmapped	dhcpd: DHCPACK on
2026-09-14 23:35:14Z	SIEM-0216	SIEM	INFORMATIONAL	VPN-GW-01	unmapped	[firewall] Inbound c
2026-09-14 23:33:44Z	SIEM-0215	SIEM	INFORMATIONAL	ENG-WS-055	unmapped	[endpoint] Software
2026-09-14 23:07:57Z	SIEM-0214	SIEM	INFORMATIONAL	MAIL-01	unmapped	[endpoint] Schedule
2026-09-14 22:58:31Z	SIEM-0213	SIEM	LOW	FIN-WS-043	unmapped	[authentication] Fail
2026-09-14 22:55:32Z	SYS-0232	SYSLOG	INFO	HR-WS-023	unmapped	sysmon: EventID=1
2026-09-14 22:50:59Z	SIEM-0212	SIEM	INFORMATIONAL	PRINT-SRV-01	unmapped	[endpoint] Endpoint
2026-09-14 22:47:28Z	GEO-0062	GEO	PH	SITE-ALPHA	unmapped	scheduled patrol: r
2026-09-14 22:46:32Z	SIEM-0211	SIEM	INFORMATIONAL	RND-WS-117	unmapped	[authentication] Fail
2026-09-14 22:40:01Z	SYS-0231	SYSLOG	INFO	RND-WS-117	unmapped	Microsoft-Windows-
2026-09-14 22:20:41Z	SYS-0230	SYSLOG	INFO	DNS-01	unmapped	kernel: [659956.445
2026-09-14 21:49:55Z	SIEM-0210	SIEM	INFORMATIONAL	VPN-GW-01	unmapped	[firewall] Inbound c
2026-09-14 21:45:45Z	SYS-0229	SYSLOG	INFO	FIN-WS-042	T1218.015 Electron Applications	sysmon: EventID=1

Alert Details: SIEM-0121 CRITICAL

Timestamp: 2026-09-14 06:00:12Z
Vendor severity: CRITICAL (verbatim, not normalised)
Asset: DC01 · dc01 · 10.50.1.10
Incident: INC-018
Queue position: 136 of 456 (incident rank)

WHY IS THIS RANKED HERE?

VENDOR SAYS: CRITICAL VS AEGIS RANKS IT: 136 of 456
SIEM severity, verbatim INC-018 · composite 6/100

1 SUPPRESSION RULE FIRED

Authorised vulnerability scanner KNOWN_SCANNER_RANGE

Scanning from the registered scanner range on the standing schedule is expected; the IDS marks every scan CRITICAL because it cannot know who is scanning.

EVIDENCE 12 of 12 alerts are scanning techniques sourced from 10.50.1.10 inside the authorised scanner range 10.50.1.0/24 (asset SCAN-01). Activity started Mon 06:00 UTC, matching the standing scan schedule: Sundays and Mondays 06:00-06:30 UTC (change CHG-STD-0802, standing approval).

SCORE BREAKDOWN

- Correlation confidence: 0.73
- Asset criticality: 1.00
- Tactic severity: 0.49
- False-positive likelihood (subtracts): 0.92

Suppression rules fired: **Authorised vulnerability scanner**

NARRATIVE

The siem vendor marked SIEM-0121 'critical'. AEGIS placed its incident INC-018 at position 136 of 456 (score 6.0/100) because 1 suppression rule(s) fired, giving a false-positive likelihood of 0.92. KNOWN_SCANNER_RANGE — Authorised vulnerability scanner: 12 of 12 alerts are scanning techniques sourced from 10.50.1.10 inside the authorised scanner range 10.50.1.0/24

What AEGIS answers

Position 136 of 456. Rule KNOWN_SCANNER_RANGE: 12 of 12 alerts are scanning techniques from 10.50.1.10 inside the authorised scanner range, Monday 06:00 UTC, matching the standing schedule. Correlation confidence 0.73, asset criticality 1.00 — and a false-positive likelihood of 0.92.

What would change its mind

A different source address, a different account, activity outside the window, or any technique beyond scanning. The rule is narrow on purpose.

IBM Bob is the analyst's interface, not a footnote

Six read-only MCP tools (Python MCP SDK, stdio)

TOOL	THE ANALYST ASKS
get_priority_queue	"What should I look at first this shift?"
explain_correlation	"Why are SIEM-0146 and SYS-0161 the same incident?"
get_bluf	"Give me the BLUF for INC-017."
why_deprioritised	"Why is critical alert SIEM-0121 ranked so low?"
search_by_technique	"Show me everything mapped to T1059."
get_incident	"Walk me through INC-014."

Built with Bob

Bob was used throughout the build: planning the module contracts, implementing the adapters and correlation signals, reviewing the tactic matrix, and driving the terminal for the pipeline runs.

watsonx.ai Granite (optional)

Rewrites the BLUF narrative from the structured evidence under a strict JSON schema. Hallucinated alert IDs are rejected; the template path is what the demo runs on.

Remove Bob and...

the analyst loses their primary conversational interface into the evidence trail — not a summary paragraph.

Measured against ground truth, not asserted

521 synthetic alerts · 8 planted scenarios (6 intrusions, 2 look-alike benign) · 466 noise · python -m src.eval.evaluate

SCENARIO	TRUTH	COVERAGE	PURITY	QUEUE POSITION	SUPPRESSION
Invoice phishing → PowerShell → C2 → DC01 NTDS → exfiltration	intrusion	1.00	1.00	1	—
UAS + RF emitter → rogue AP → OT VLAN → PLC write (SITE-ALPHA)	intrusion	0.86	1.00	2	—
VPN password spray → account takeover → AD/share enumeration	intrusion	1.00	1.00	3	—
CVE-2023-22518 → web shell → cron → reverse shell	intrusion	1.00	1.00	4	—
Shadow-copy deletion → masquerading → mass encryption	intrusion	1.00	1.00	5	—
Insider staging → personal cloud upload (ambiguous)	intrusion	1.00	1.00	6	—

Results & impact

Sysadmin rollout inside change window benign 0.75 1.00 117

MAINTENANCE_WINDOW

Team Blue Vector — and where this goes

MEMBER	BUILT
Het Khatusuriya (lead)	Corpus & ground truth, IOC extraction, tactic ordering matrix, suppression rules, asset inventory
Manan Modi	ATT&CK mapping, correlation graph & signal weights, BLUF generation, evaluation harness
Mohammad Safik	Analyst console: queue, correlation graph, kill-chain view, BLUF panel, screenshots
Samarth Bhalala	Canonical schema, four ingest adapters, SQLite store, FastAPI, MCP server for Bob

Beyond the hackathon

- Streaming ingestion with incremental graph updates
- Suppression knowledge from the CMDB and change system, not code
- Analyst feedback loop to recalibrate signal weights per enclave
- Bob as the first-line triage assistant on the watch floor

`github.com/het-khatusuriya/bob-ai-hackathon-blue-vector`

`python scripts/demo.py → http://localhost:8000`